MULE PULSE

Presented By Team HAHA

PROBLEM STATEMENT

Malaysia's current counter-fraud response excels at reactive tracing, but post-scam fund recovery has become a critical failure point due to accelerating mule networks.

RM 5.62B

Cumulative Fraud Losses (2023-2025)
RM1.28B('23)→RM1.57B('24)→RM2.77B('25)

87,209

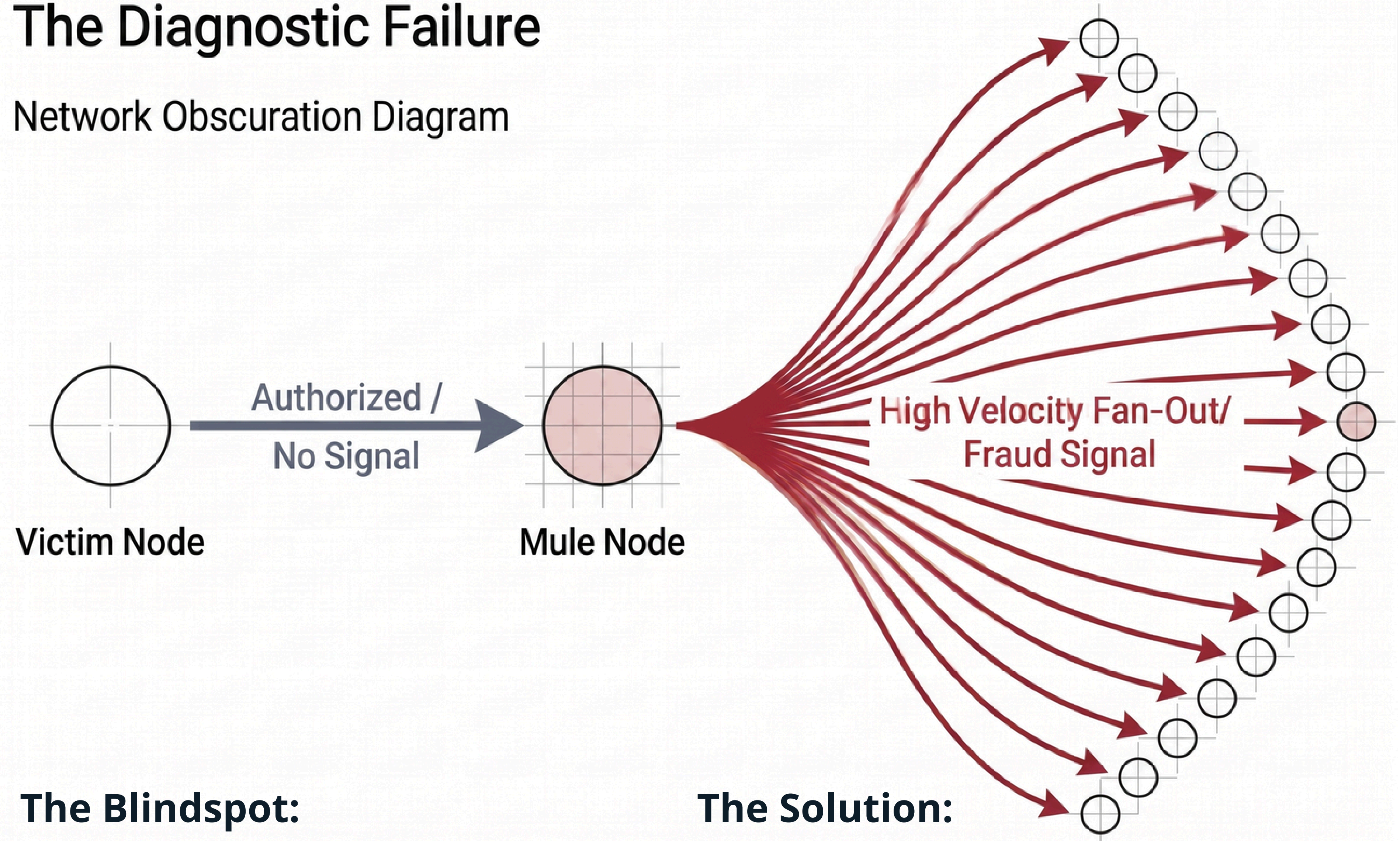
Active Mule Accounts in 2025.
A 70% YoY acceleration from 51,302 in 2024.

6%

The Recovery Rate.
Only RM34M recovered from RM542M reported losses in 2025,

The Diagnostic Failure

Network Obscuration Diagram



The Blindspot:
In 95% of Malaysian scams, victims knowingly authorize transfers. The outbound transaction appears legitimate to isolated, bank-level rule engines.

The Solution:
The true fraud signal resides on the receiving side, hidden within the structure, velocity and fan-out behaviour of the mule network

KEY FEATURES

Layer 4: AI Investigation Agent

- Auto-assembles case files with plain-language, explainable narratives.
- Recommends actions; explicitly requires human analyst approval.

Layer 3: Mule Risk Scoring Model

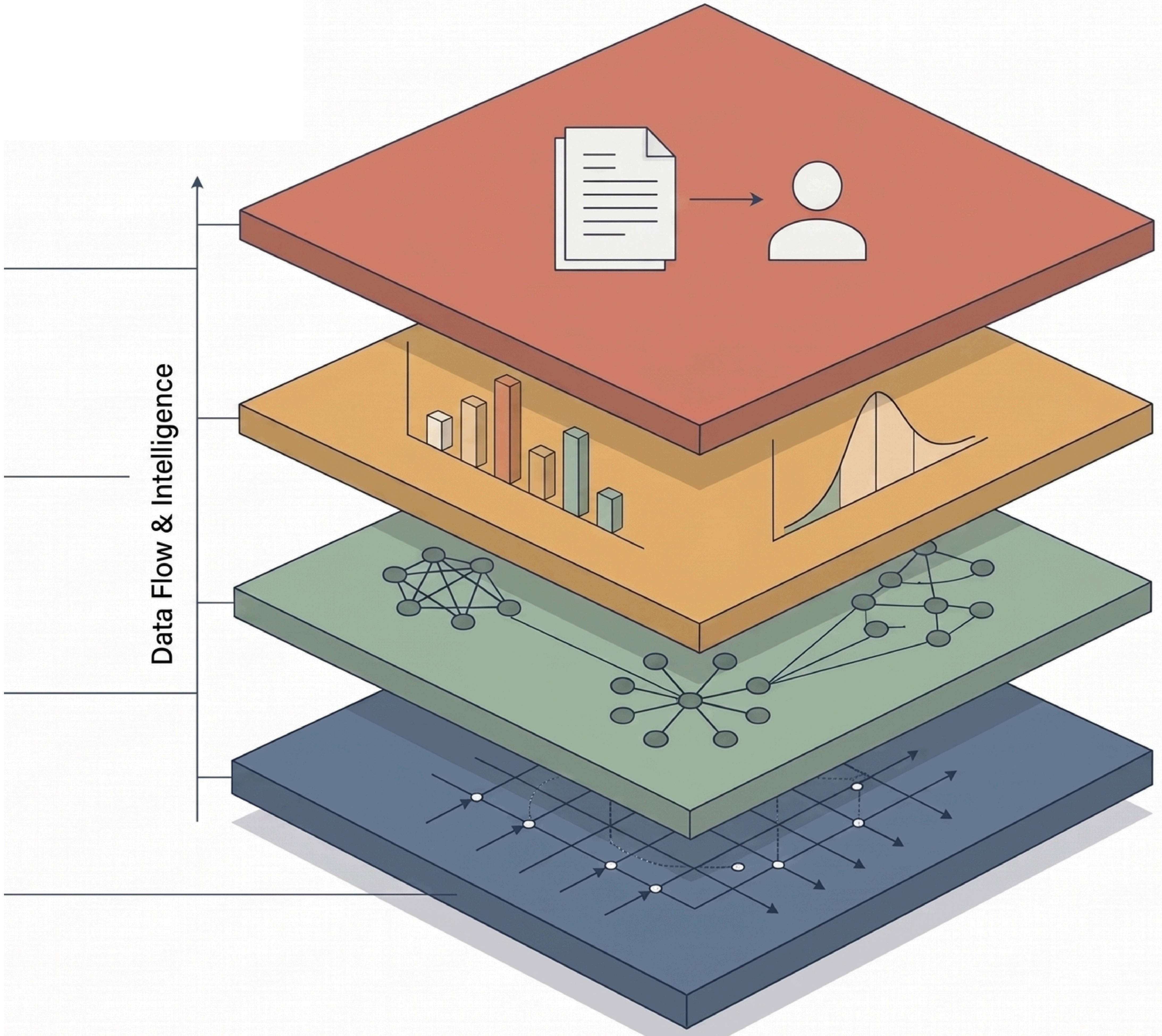
- Machine learning combines grpah features to output calibrated risk scores.
- Evaluates entire clusters simultaneously with fully tunable thresholds.

Layer 2: Suspicious Pattern Extraction

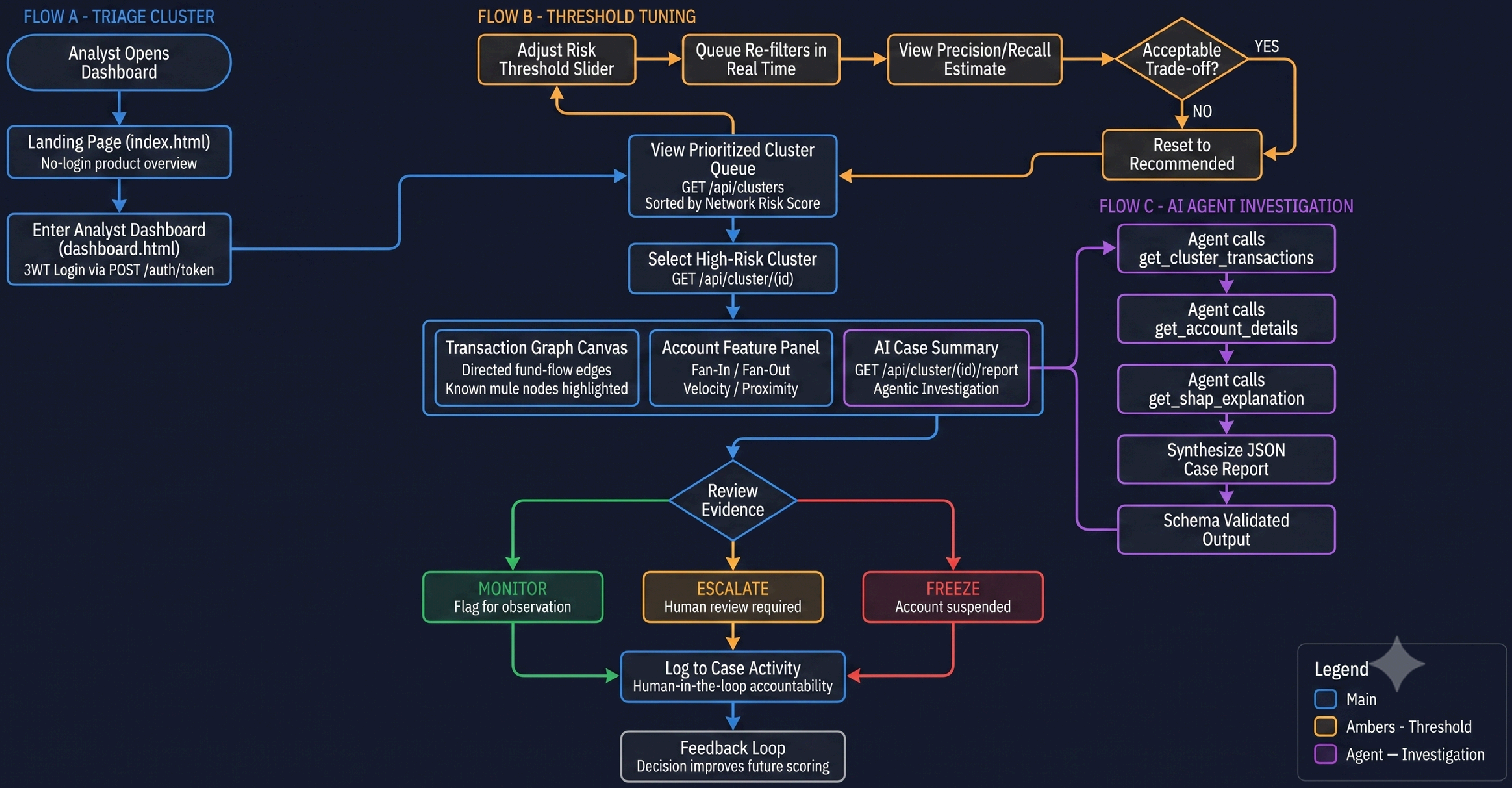
- Surfaces structural signals: Fan-in, Fan-out, and Pass-through velocity
- Executes community detection to surface entire coordinated rings.

Layer 1: Transaction Graph Engine

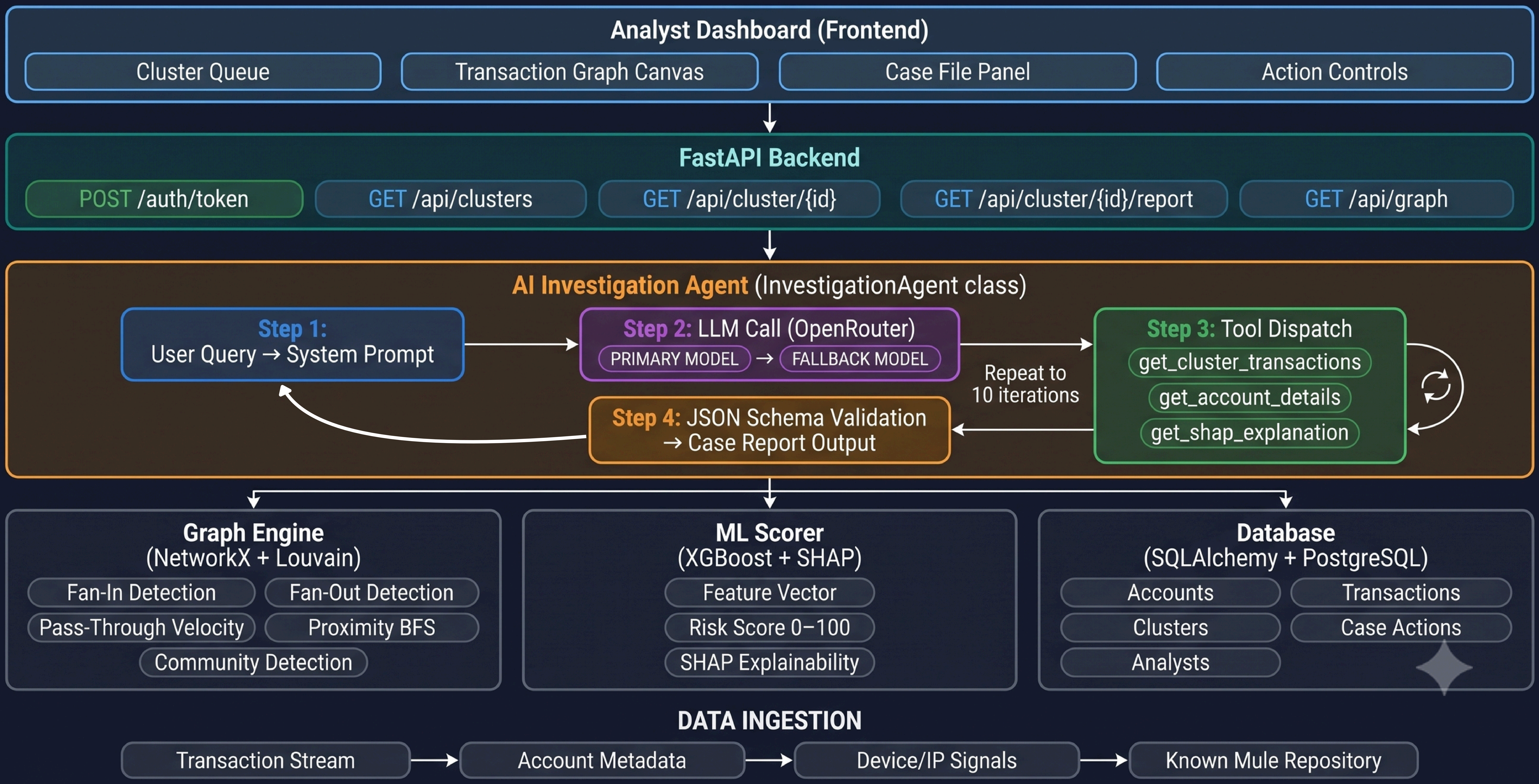
- Ingests streams to build directed, time-aware graphs in near-real-time.
- Edges weighted by velocity and time-decay.

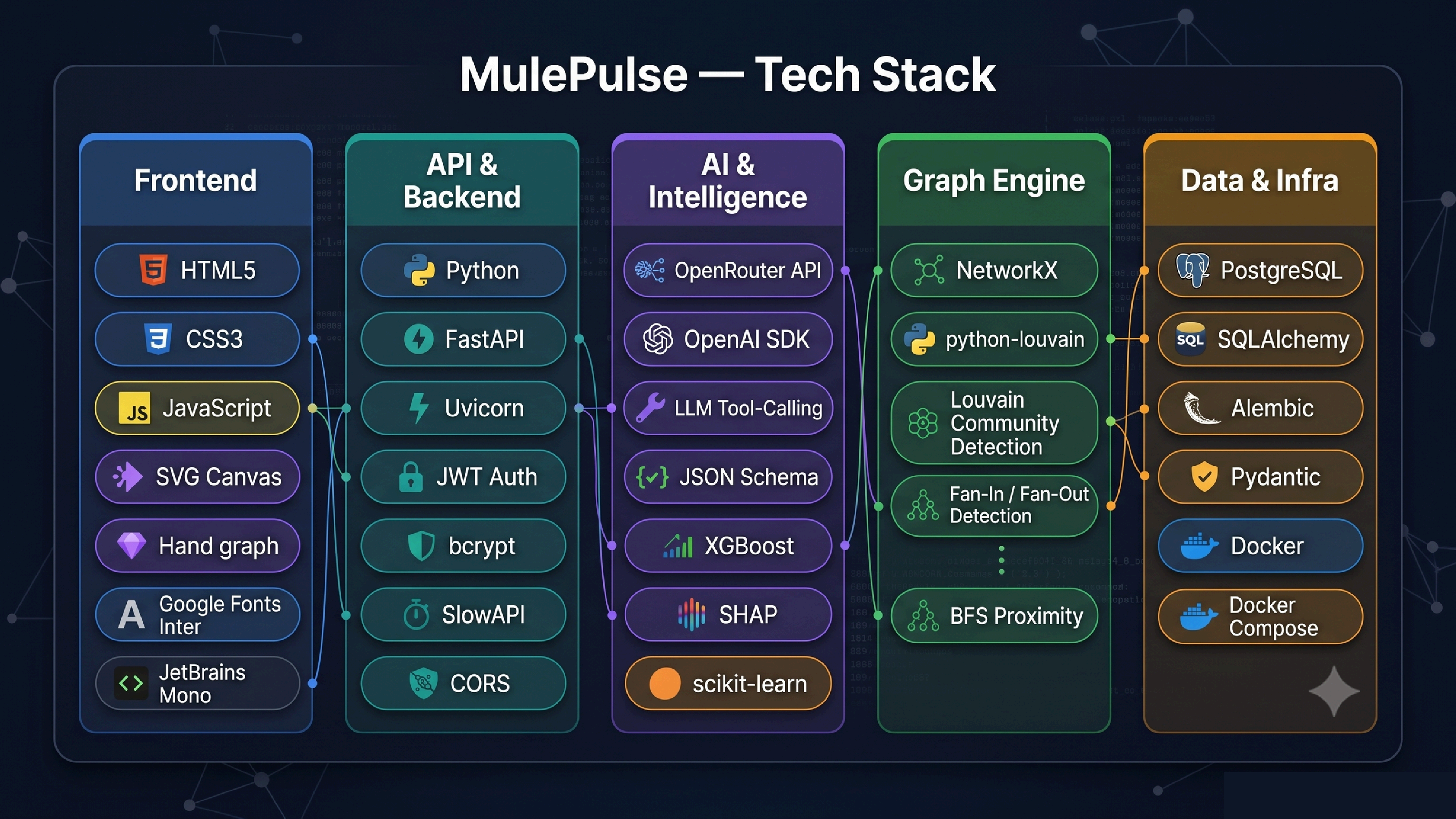


MulePulse — Analyst User Flow



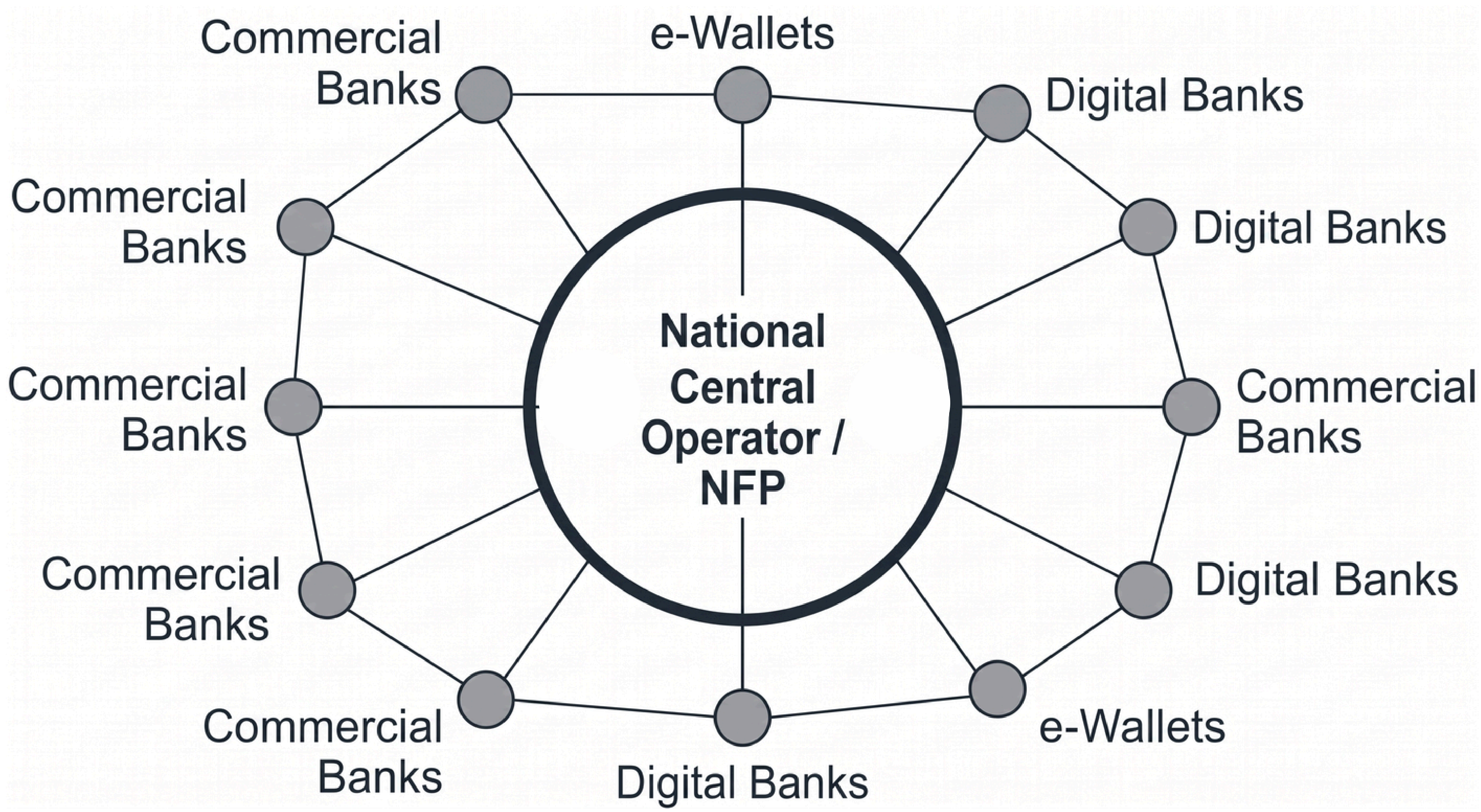
MulePulse — Agent Framework Architecture




MulePulse — Tech Stack
Frontend
HTML5
CSS3
JavaScript
SVG Canvas
Hand graph
Google Fonts Inter
JetBrains Mono
API & Backend
Python
FastAPI
Uvicorn
JWT Auth
bcrypt
SlowAPI
CORS
AI & Intelligence
OpenRouter API
OpenAI SDK
LLM Tool-Calling
JSON Schema
XGBoost
SHAP
scikit-learn
Graph Engine
NetworkX
python-louvain
Louvain Community Detection
Fan-In / Fan-Out Detection
BFS Proximity
Data & Infra
PostgreSQL
SQLAlchemy
Alembic
Pydantic
Docker
Docker Compose

TARGET MARKET

MulePulse serves a single buyer: the national Central Operator (i.e. Paynet), delivering simultaneous, cross-institutional protection for the entire financial market.



The Central Hub Advantage

Mule syndicates deliberately fragment operations across multiple competing institutions.

By deploying at the Central Operator Tier, MulePulse gains lawful, cross-institutional visibility.

One procurement translates into system-wide protection for every participating bank, e-wallet and consumer.

Individual FIs act as supporting data partners. MulePulse acts as the **missing pre-emptive discovery layer** atop the exisitng National Fraud Portal stack.

IMPLEMENTATION PLAN

Phase 1

Proof of Value on Historical Data (3–4 months)

- Backtest on the operator's historical data and confirmed-mule repository
- NFP mule database powers detection as training labels, graph anchors, and backtest ground truth
- Quantify the lead time gained ahead of victim reports; engage early high-volume reference partners

Phase 2

Shadow Mode Evaluation on Live Data (4–6 months)

- Score live transactions in parallel, taking no action — validates accuracy, false positives, and timing
- Existing fraud-response process stays fully authoritative throughout
- Broaden participation to highest-volume institutions for maximum early national coverage

Phase 3

Production Deployment & Analyst Workflow (6 months)

- AI Investigation Agent delivers explainable case files to operator-level case officers
- Every irreversible action requires human approval — never autonomous, fully logged for audit
- Extend participation to mid-sized and smaller institutions for full-breadth coverage

Phase 4

Continuous Learning & Operational Maturity (ongoing)

- Analyst decisions retrain the model — sharper precision, fewer repeat false positives
- Confirmed mules enrich the NFP database — operator's ground truth and model improve together
- Full national coverage reached; network effect matures as visibility completes

De-risked by design: institutions onboard through existing NFP connections, phase by phase

COMPLIANCE

OPERATIONAL GOVERNANCE (BNM)

Positioned within the Paynet/NFP operator tier under BNM oversight.
Inherits the lawful basis used to share mule and fraud data.
Rigorously conforms to BNM’s Risk Management in Technology standards.

DATA PROTECTION UNDER PDPA

Processing permitted under the crime-prevention and detection exemption (Section 13, PDPA 2010).
Built to meet the PDPA's 2024 security, breach-notification and DPO obligations.